ASSIGNMENT NO. 2

```java
package Hi;
import java.io.*;
import java.util.*;
public class Pass2 {
  public static void main(String args[]) throws Exception {
    // Read symTable
    HashMap<Integer, Integer> symtab = new HashMap<>();
    BufferedReader br = new BufferedReader(new FileReader("symtab.txt"));
    String line;
    while ((line = br.readLine()) != null) {
      String parts[] = line.trim().split("\\s+");
      int index = Integer.parseInt(parts[0]);
      int address = Integer.parseInt(parts[2]);
      symtab.put(index, address);
    }
    br.close();
    // Read lit Table
    HashMap<Integer, Integer> littab = new HashMap<>();
    br = new BufferedReader(new FileReader("littab.txt"));
    while ((line = br.readLine()) != null) {
      String parts[] = line.trim().split("\\s+");
      int index = Integer.parseInt(parts[0]);
      int address = Integer.parseInt(parts[2]);
      littab.put(index, address);
    }
    br.close();
    // Read IC
    br = new BufferedReader(new FileReader("op.txt"));
    // Create Machine Code
    BufferedWriter bw = new BufferedWriter(new FileWriter("machinecode.txt"));
    System.out.println("MACHINE CODE");
    System.out.println("----------------");
    while ((line = br.readLine()) != null) {
      line = line.trim();
      if (line.length() == 0)
        continue;

      if (line.contains("(AD,01)") || line.contains("(AD,02)")) {
        continue;
      }
      String parts[] = line.split("\\s+");
      // IMPERATIVE STATEMENT
      if (line.contains("(IS")) {
        String opcode = parts[1].replace("(IS,", "").replace(")", "");
```

```java
            String register = parts[2].replace("(RG,", "").replace(")", "");


            int address = 0;
            // Symbol
            if (parts[3].startsWith("(S")) {
                int index = Integer.parseInt(
                    parts[3].replace("(S,", "").replace(")", "")


                );
                address = symtab.get(index);
            }
            // Literal
            else if (parts[3].startsWith("(L")) {
                int index = Integer.parseInt(
                    parts[3].replace("(L,", "").replace(")", "")


                );
                address = littab.get(index);
            }
            String machineCode = opcode + " " + register + " " + address;
            System.out.println(machineCode);
            bw.write(machineCode);
            bw.newLine();
        }
        // DS
        else if (line.contains("(DL,01)")) {
            String machineCode = "00 00 000";
            System.out.println(machineCode);
            bw.write(machineCode);
            bw.newLine();
        }
        // DC
        else if (line.contains("(DL,02)")) {
            String constant = parts[2].replace("(C,", "").replace(")", "");


            String machineCode = "00 00 " + constant;
            System.out.println(machineCode);
            bw.write(machineCode);
            bw.newLine();
        }
    }
    br.close();
    bw.close();
    System.out.println("----------------");
    System.out.println("Machine code generated successfully.");
```

```
  }
}
```

machinecode.txt   op.txt   Re

```
(AD,01) (C,100)
100 (IS,05) (RG,01) (L,0)
101 (IS,04) (RG,02) (S,0)
102 (IS,02) (RG,03) (S,1)
103 (DL,01) (C,1)
104 (DL,02) (C,2)
(AD,02)
105 (DL,02) (C,5)
```

machinecode.txt   littab.txt

```
0 ='5' 105
```

machinecode.txt   symtab.txt

```
0 X 103
1 Y 104
```

machinecode.txt

```
05 01 105
04 02 103
02 03 104
00 00 000
00 00 2
00 00 5
```

```
<terminated> Pass2 [Java Application] C:\Users\Hp\
MACHINE CODE
----------------
05 01 105
04 02 103
02 03 104
00 00 000
00 00 2
00 00 5
----------------
Machine code generated successfully.
```